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NPS/001/001 Technical Specification for Wood Poles and Associated Products for Overhead Lines

1. Purpose

This specification details the requirements of CE Electric UK for the supply of wood poles and associated products for use on the overhead distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/004/005	2.0	Specification for Poles and Associated Products for Overhead Lines
MS-O-29	1.0	Specification for Wood Poles for Overhead Lines up to 132kV
NPS/001/001	1.0	Specification for Wood Poles and Associated Products for Overhead Lines

2 Scope

This specification details the required characteristics of wood poles and associated timber required for the construction and maintenance of LV, 11kV, 20kV, 33kV, 66kV and 132kV overhead lines for use on the CE Electric UK distribution network.

3 Specification

3.1 General

Poles shall be selected, fabricated and treated in accordance with ENA TS 43-88 Issue 4 1993 except where detailed otherwise within this specification. **All suppliers must complete and return the self certification documents contained in Appendix A of this specification.**

Timber strengths shall be determined in accordance with the technical requirements expressed in product standards BS EN 12509 and BS EN 12511. The adoption of these standards has introduced new methods of tests and statistical analysis in terms of the forest population. However, these product standards are currently under review.

Until the new harmonised standard with its associated technical revisions is adopted it is unlikely that new statistical data will be available.

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Until further information is available poles to this specification shall be supplied with the following 'characteristic values'.

10,054 Nmm² for modulus of elasticity and 53.35Nmm² for modulus of rupture should be considered as representative values for a population defined as mainland Europe north of 60° latitude.

For any species and population the following performance requirements shall be declared; length, and the mean and characteristic bending strength and the mean and the 5% fractile characteristic modulus of elasticity combined with the minimum nominal diameter at 1.5 m from the butt and the minimum diameter at the tip.

3.2 Clause by Clause Variations to ENA TS 43-88 Part 1 Issue 4 1993

Clause 4.1 Species for Poles

Redwood (Pinus Sylvestris)
Scots Pine (Pinus Sylvestris)
Douglas fir (Pseudotsuga Menziesii)
Corsican Pine (Pinus Nigra)

The preferred species are Redwood and Scots Pine. Other suitable species and grading requirements may be acceptable as agreed between the Buyer and the Supplier.

This Specification refers to the following species of imported and homegrown softwood timber for use as poles:

TABLE 1

Species	Origin	Sapwood Permeability
Redwood (Pinus Sylvestris)	Imported or Homegrown	permeable
Douglas Fir (Pseudotsuga Menziesii)	Imported or Homegrown	moderately resistant
Corsican Pine (Pinus Nigra)	Homegrown	permeable

Clause 4.3 Felling

The trees shall be felled during the period when the rising sap is low. If necessary, the trees can be felled when the sap is high provided necessary precautions are taken to prevent pre-treatment decay or attack by insects. These measures shall be agreed between Supplier and Buyer.

The timber shall not be accepted if the period between felling and delivery at the Suppliers Works exceeds 20 months, or such a period agreed between the Supplier and Buyer. Homegrown timber shall be removed from the forest within one month of felling.

Clause 5.1 Size

The size of a pole shall be specified by the overall length, the nominal diameter at 1.5 metres from the butt and the nominal diameter at the tip. The poles shall comply with the minimum diameters and lengths quoted in Table 2 below after dressing and be of uniform taper or as otherwise specified.

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Table 2 - Commonly used sizes for wood poles

Nominal Length	Light Poles		Medium Poles		Stout Poles		Extra Stout Poles	
	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt
m	mm	mm	mm	mm	mm	mm	mm	mm
6	125	150						
7	125	160	140	200				
8	125	170	145	210				
8.5	125	180	150	215	190	265		
9	125	180	150	220	190	275	230	295
9.5	125	185	150	225	190	280		
10	125	185	150	230	190	285	230	305
10.5	125	190	150	235	190	290		
11	125	195	150	240	190	295	245	320
11.5	125	200	150	245	190	300		
12	125	200	150	250	190	305	245	335
13	130	210	160	260	195	320	245	350
14			160	275	195	335	245	365
15			165	290	195	350	245	375
16			170	305	200	365	250	390
17			180	320	200	375	250	415
18			180	330	200	390	250	415
20			180	360	200	415	260	435
22			190	380	200	435	260	470
24					200	470		

Clause 7.4 Pole Fabrication and Scarfing

All pole fabrication shall take place prior to preservative treatment. Appendix 1 provides details of the required pole height and grade together with its associated fabrication requirements.

The CE Electric UK has a requirement for a range of additional pole fabrications, to cater for in house company design standards or to support special pole mounted plant. The company will submit copies of these pole fabrication drawings at the time of order placement.

All 'H' and Rutter poles shall be supplied as matched pairs, additionally matched pairs of poles should be supplied so that they are readily identifiable from similar pairs in the delivery.

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Clause 7.5 Pole Marking

ENA TS 43-88 refers to Appendix C for pole marking details, out of the list shown on Appendix C the following items shall be marked on all poles

Length and Class	Required
Gouge Mark (3 meters from pole butt)	Required
Species	Required
Country of Origin	Required
Year Of Treatment	Required
Type of Preservative Treatment	Required
Charge No.	Required

Note – an additional 3 meter gouge shall be marked at 180° to the one shown in ENA TS 43-88 Part 1: Appendix C.

Clause 8.0 Preservative Treatment (Creosote – all poles fabricated for HV use)

Clause 8.1 General

The following treatment description is pertinent to home grown and imported Pinus Sylvestris only using the Rueping empty cell process. Creosoting shall be carried out to BS 144: Part 2.

Clause 8.4 Creosote Quality

The creosote quality shall be to Type B or Type C oil of BS EN 13991. It will have a benzo-a-pyrene content of less than 0.005% by mass (50 ppm), and water extractable phenols at a concentration of less than 3% by mass.

Clause 8.5 Quality Control of Preservative Treatment

Clause 8.5.1 General

Preservative treatment shall be defined in terms of depth of lateral penetration of preservative into the treated pole and retention of the preservative within that treated zone. For the purpose of verifying compliance with this Standard, a charge will be considered.

In order to ensure consistent quality in the preservative treatment of wood poles, testing will be carried out on every charge of wood poles.

Clause 8.5.2 Creosote Penetration

The sapwood shall be completely impregnated with creosote to penetration class P8 in accordance with the requirements of BS EN 350 Part 1, which shall be verified by means of test borings and subsequent testing of these borings using a sapwood / heartwood reagent.

No poles shall be released for delivery until the results of the test borings are satisfactory.

The preservative penetration of a charge will be evaluated by boring a number of samples in accordance with Table 3. In the event of a charge of preservative treated poles being rejected the charge or batch may be retreated and reassessed. Alternatively each component found to comply may be accepted.

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If sample size meets or exceeds the lot or batch size, carry out 100% inspection.

Table 3

Number of Samples to be Taken and Maximum Number Allowed to Fail, to Requirements of BS 6001 (General Inspection Level II, AQL, 0.65)		
Batch Size	Sample Size	Allowable Failure
1-15	3	0
16-25	5	0
26-50	8	0
51-90	13	0
91-150	20	0
151-280	32	0
281-500	50	1
501-1200	80	1

Clause 8.5.3 Creosote Retention

The net retention shall be 185kg/m³ with tolerance range to be – 10% to unlimited retention above this value in the treatable zone, or as agreed between the Buyer and Supplier. The value of 185kg/m³ is based on a sapwood ratio of 62% and is equivalent to a sapwood treatment of 115kg/m³ expressed on the total wood volume determined from average calliper volumes stated in ENA TS 43-88 Appendix A1. Direct testing of this retention will be achieved by quantitative chemical analysis of the complete analytical zone separated from each boring in 8.5.2 combined into a single sample.

Indirect Testing, such as the volumetric method described in ENA TS 43-88 Appendix E, may only be used to determine the quality of the preservative treatment on a charge basis where a safe relation has been established between both the penetration, and retention requirements of 8.5.2 and 8.5.3 and other properties associated with the treated pole.

An independent third party will verify the validity of the safe relation at least every six months by carrying out direct and indirect testing on the same charge of treated poles.

Clause 8.6 Bleeding poles

Bleeding poles must not be dispatched, and poles that subsequently bleed before erection shall by arrangement be returned for re-treatment. The re-treatment of bleeding poles shall be carried out on a charge comprising bleeding poles only.

Clause 8.7 Maintenance of Pole Condition

After treatment all stored poles shall be stacked as in ENA TS 43-88 Clause 6.1.

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Clause 9.0 Preservative Treatment (Water Soluble preservatives) – (All poles fabricated for LV use)

Clause 9.1 General

Pinus Sylvestris and Pinus Nigra shall be treated using the high pressure process specified in BS 8417 clause 4.3.3

Clause 9.3 Water soluble Preservative

The following formulations of water soluble preservatives may be used.

- Celcure AC-500 – (Poles to be marked with AC)
- Tanalith E 3494 – (Poles to be marked with E3494)

The critical value of these preservatives shall be determined in accordance with BS EN 599-1 appropriate to hazard class 4 in EN 335-1.

Clause 9.4 Water Soluble Preservative Loading and Tests

The treatment cycle shall be such as to give complete penetration of the sapwood to penetration class P8. The minimum retention shall be appropriate to a 30 year desired service life specified in Table 9 of BS 8417. Preservative retention may be determined using indirect methods. The volume of the sapwood in the charge shall be determined from the average calliper cubes in ENA TS 43-88 Appendix A1. The calculations shall assume a sapwood ratio of 62%.

Clause 11.0 Stay and brace blocks

Wood Blocks shall be as specified in ENA TS 43-91 and 43-88 and treated with creosote or water soluble preservatives.

For permeable species, full sapwood penetration P8 is required, for resistant species P7 is required in any incised zone and P5 required elsewhere in accordance with the requirements of BS EN 350 Part 1.

Clause 13.0 Quality Requirements

Clause 13.1 General

All suppliers shall be able to comply with the full requirements of ENA TS 43-88 specification; the additional clauses listed in this specification and shall have a minimum third party accreditation to EN ISO 9001.

The pole supplier shall have in operation a quality system and documentation shown in Section 14 of this Specification. Inspectors representing the Buyer may visit the Suppliers works at random to verify quality procedures are maintained at all stages.

Clause 13.2 Specific

In addition to 13.1 above suppliers shall have in operation the following:

1. A system to record that all operators and those persons making quality inspections, have been certified as having received formal training and adequate experience in the individual tasks they are expected to perform.

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2. A recorded certification system for measuring all instruments to verify their accuracy is consistent with manufacturers recommended use and any periodic maintenance and recalibration
3. the results of inspections, tests or assessments requiring action shall be recorded, as shall any actions taken when control values or criteria are not met

Clause 14 Suppliers Documentation

Clause 14.1 General

The Suppliers records shall contain the following information, and shall be kept for at least 7 years.

The Clause numbers refer to the relevant Clauses within this Specification.

Clause 14.2 Quality Control of Poles

- | | | |
|-------|---|-------------------|
| i) | Job or order number and customer's name | (7.5) |
| ii) | Timber Species | (4, 11, and 12) |
| iii) | Timber Species Population | (5.3) |
| iv) | Date of Felling | (4.3) |
| v) | Date of arrival at Suppliers Works | (4.3) |
| vi) | Stacking conforms to Specification | (6.1) |
| vii) | Grading | (5.1 and 5.2) |
| viii) | Date and working shift | (5.1.2) |
| ix) | Grader's name | (5.1.2) |
| x) | Checks on moisture content have been made as required | (6.2, 11, and 12) |
| xi) | Conformity of fabrication | (7, 11, and 12) |

Clause 14.3 Quality control of preservative treatment

Clause 14.3.1 Creosoted Poles

- | | | |
|------|---|---------------|
| i) | Treatment cycle conforms to BS 144, Part 2 or as agreed | (8.1) |
| ii) | Creosote conformity to EN 13991 Type B | (8.4) |
| iii) | Penetration and retention of preservative treatment | (8.5) |
| iv) | Incidents of bleeding poles | (8.6) |
| v) | Poles stacked to specification | (6.1 and 8.7) |
| vi) | Creosote Certificates (no Clause) | |

Clause 14.3.2 Re Treatment (Creosote)

- | | | |
|------|---|-----------------|
| i) | Moisture content | (10.1) |
| ii) | Checks on scarfing details | (7.4) |
| iii) | Creosote Penetration and Retention | (8.5) |
| iv) | Penetration and retention of preservative treatment | (8.5) |
| v) | Additional net retention as required | (10.1 and 10.2) |

Clause 14.3.3 Water Soluble Preservatives

- | | | |
|------|---|---------------|
| i) | Composition of water soluble preservative | (9.3) |
| ii) | Treatment cycle | (9.1) |
| iii) | Penetration and retention of preservative treatment | (9.4 and 9.5) |
| iv) | Poles stacked to specification | (6.1 and 9.6) |

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Clause 14.3.4 Re-treatment (Water Soluble Preservatives)

- i) Poles not meeting the required retention level and also (9.7 and 10.3)
- ii) Ones pre-treated before fabrication are re treated

Clause 14.3.5 Stay Blocks, Brace Blocks and Ancillary Timber

- i) Creosote quality (11.1)
- ii) Stacking in cylinder (11.1)
- iii) Percent of total charge in cylinder (11.1)
- iv) Penetration checks (11.1)

4 References

Reference	Title
ENA TS 43-88 Parts 1 & 2	"Selection and Treatment of Wood Poles and Associated Timber for Overhead Lines" Part 1 – Technical Requirements: 1993 Part 2 – Essential Quality Requirements: 1992
BS 1990: Part 1 – 1984	"Wood Poles for Overhead Power and Telecommunication Lines"
BS 8417	"Preservation of timber - recommendations"
ENA TS 43-91	"Stay Strands and Stay Fittings for Overhead Lines"
BS EN 13991	Derivatives from Coal pyrolysis – Coal tar base oils: creosotes – Specifications and test methods
BS 144	Methods for timber treatment
BS 4978	Specification for visual strength grading of wood poles
BS EN 252	Field test method for determining the relative protective effectiveness of a wood preservative
EN 335 – Part 1	Hazard classes of wood and wood based products against biological attack – Part 1 classification of hazard classes
BS EN 351 Part 1	Durability of wood and wood based products - Preservative treated solid wood – Part 1. Classification of preservative penetration and retention.
BS EN 351 Part 2	Durability of wood and wood based products - Preservative treated solid wood – Part 2. Guidance for the sampling for the analysis of preservative treated wood.

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BS EN 12465	Wood poles for overhead lines – Durability requirements
BS EN 12479	Wood poles for overhead lines – Sizes – Methods of measurement and permissible deviations.
BS EN 12509	Wood poles for overhead lines – Test methods – Determination of modulus of elasticity, bending strength, density and moisture content.
BS EN 12510	Wood poles for overhead lines – Strength grading criteria.
BS EN 12511	Wood poles for overhead lines – Determination of characteristic values.

Reference	Title
BS EN 599 Part 1	Durability of wood and wood based products. Performance of wood preservatives determined by biological tests – Part 1: Specification according to hazard class.

5 Definitions

Term	Definition

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G.Hammel	Overhead Line Standards Manager	G.Hammel	26/11/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M. Nicholson	Head of System Strategy	M. Nicholson	26/11/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
S.Salkeld	Standards Engineer	S.Salkeld	26/11/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S. Johnson	DBD Administrator	S. Johnson	14/12/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C. Holdsworth	Standards & Assessment Manager	C. Holdsworth	27/11/07

6.6 Authorisation

Approval is given for the content of this document

		Sign	Date
C. Morgan	Head of Procurement	C. Morgan	28/11/07

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Appendix A

This conformance declaration is to be completed.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'EATS' as appropriate

Manufacturer:

Product Reference :

Name:

Signature:

Date:

TABLE 1

BS EN 12510			EATS 43 – 88 Part 1			Remarks
Clause / Sub-clause	Requirement	Conformance code	Clause / Sub-clause	Requirement	Conformance code	
5.2	Species ¹		4.1	Species for Poles		
			4.2	Timbers other than for Poles		
5.3.1	Felling		4.3	Tree Felling ²		
6	Visual Grading ³		5	Requirements ⁴		

¹ Species within EN 12510 reflect broad European practice that may not be acceptable in the UK. The permitted species within EATS 43-88 reflects historical UK practice and represents a condensed list of the permitted species within EN 12510.

² EATS 43-88 places additional requirements on the maximum period between felling and delivery to supplier's works.

³ Subject to the additional requirements of Clause 5.1.3 of EATS 43-88 [Table3 of this Appendix]

⁴ EATS 43-88 sets limit deviations for the visual grading rules laid down in BS EN 12510

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BS EN 12510			EATS 43 – 88 Part 1			Remarks
Clause / Sub-clause	Requirement	Conformance code	Clause / Sub-clause	Requirement	Conformance code	
5.3.3	Storage of Untreated Wood		6	Seasoning		

TABLE 2

BS EN 12465			EATS 43-88 Part 1			Remarks
Clause / Sub-clause	Requirement	Conformance code	Clause / Sub-clause	Requirement	Conformance code	
5.1.1	Requirements for Raw Materials - Wood Poles		8.2	Condition of Timber (prior to treatment with Creosote)		
5.2.1	General ⁵		8.3	Creosote Treatment Cycle		
5.1.2	Wood Preservative ⁶		8.4	Creosote Quality		
5.2.2/5.2.3/5.2.4	Preservative Treated Wood Poles		8.5	Quality Control of Preservative Treatment		
7	Sampling					
			8.5.3	Creosote Retention		
			8.6	Bleeding Poles (Creosote)		
			8.7	Maintenance of Pole Condition (Creosote)		

⁵ Treatment Cycle appropriate to Creosote is not defined in BS EN 12465 but the principles of Clause 5.2.1 apply.

⁶ Subject to the requirements of BS EN 13991 for a Type B oil.

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BS EN 12465			EATS 43-88 Part 1			
Clause / Sub-clause	Requirement	Confor-mance code	Clause / Sub-clause	Requirement	Confor-mance code	Remarks
5.2.1	General ⁷		9.1	General – Treatment process		
5.1.1	Requirements for Raw Materials - Wood Poles		9.2	Condition of Timber (prior to treatment with water soluble preservatives)		
5.1.2	Wood Preservative ⁸		9.3	Types of Water Soluble Preservative		
5.2.2/5.2.3 /5.2.4	Preservative Treated Wood Poles		9.4	Water Soluble Preservative Loading and Tests		
			9.6	Drying and Maintenance of Pole Condition (Water Soluble Preservative)		
6	Factory Production and Control					

TABLE 3

BS EN 12479			EATS 43-88 Part 1			
Clause / Sub-clause	Requirement	Confor-mance code	Clause / Sub-clause	Requirement	Confor-mance code	Remarks
5	Requirements for Measurement and Limit Deviations		5.1.3	Size		

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TABLE 4

<u>BS EN 13991</u>			EATS 41-36 - Part 2			Remarks
Clause / Sub-clause	Requirement	Conformance code	Clause / Sub-clause	Requirement	Conformance code	
5	Specifications and test Methods		8.4	Creosote Quality		

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Appendix B - Pole Fabrication and Arrangement Schedule

Item Ref.	Grade	Single Or 'H'	Length (m)	Stock Code	Pole Fabrication Drawing Number	Preservative	Estimated Annual Usage
LV Poles to ENA TS 43-12 for ABC & ENA TS 43-30 for Open Wire Construction							
1	Med	Single	9.0	237435	1091193301 sht3	CAA	
2	Med	Single	10.0	237454	1091193301 sht3	CAA	
3	Med	Single	11.0	237469	1091193301 sht3	CAA	
4	Med	Single	12.0	212506	1091193301 sht3	CAA	
5	Stout	Single	9.0	212944	1091193301 sht3	CAA	
6	Stout	Single	10.0	213157	1091193301 sht3	CAA	
7	Stout	Single	11.0	213538	1091193301 sht3	CAA	
8	Stout	Single	12.0	213843	1091193301 sht3	CAA	
9	Stout	Single	13.0	237470	1091193301 sht3	CAA	
10	Stout	Single	14.0	237471	1091193301 sht3	CAA	
HV Poles to ENA TS 43-40 - Medium Grade							
11	Med	Single	9.0	237331	1000434001	Creosote	
12	Med	Single	10.0	237346	1000434001	Creosote	
13	Med	Single	11.0	237350	1000434001	Creosote	
14	Med	Single	12.0	237365	1000434001	Creosote	
HV Poles to ENA TS 43-40 - Stout Grade							
15	Stout	Single	10.0	237384	1000434001	Creosote	
16	Stout	Single	11.0	237399	1000434001	Creosote	
17	Stout	Single	12.0	237401	1000434001	Creosote	
18	Stout	Single	13.0	237416	1000434001	Creosote	
19	Stout	Single	14.0	237420	1000434001	Creosote	
HV Poles to ENA TS 43-40 - Stout 'H' Grade							
20	Stout	'H'	10.0	213116	434002	Creosote	
21	Stout	'H'	11.0	237492	1000434002 sht 2	Creosote	
22	Stout	'H'	12.0	213975	434002	Creosote	
23	Stout	'H'	13.0	214189	434002	Creosote	
24	Stout	'H'	14.0	214379	434002	Creosote	

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Item Ref	Grade	Single Or 'H'	Length (m)	Stock Code	Pole Fabrication Drawing Number	Preservative	Estimated Annual Usage
Stout Grade Single poles with fabrication drawing as stated at time of order							
25	Stout	Single	9.0	233573	Various	Creosote	
26	Stout	Single	10.0	232713	Various	Creosote	
27	Stout	Single	11.0	232605	Various	Creosote	
28	Stout	Single	12.0	232639	Various	Creosote	
29	Stout	Single	13.0	232643	Various	Creosote	
30	Stout	Single	14.0	232658	Various	Creosote	
31	Stout	Single	15.0	232662	Various	Creosote	
32	Stout	Single	16.0	232728	Various	Creosote	
33	Stout	Single	17.0	232677	Various	Creosote	
34	Stout	Single	18.0	232681	Various	Creosote	
35	Stout	Single	20.0	232681	Various	Creosote	
36	Stout	Single	22.0	214890	Various	Creosote	
37	Stout	Single	24.0	215004	Various	Creosote	
Extra Stout Grade Single poles with fabrication drawing as stated at time of order							
38	Extra Stout	Single	10.0	232810	Various	Creosote	
39	Extra Stout	Single	11.0	232811	Various	Creosote	
40	Extra Stout	Single	12.0	232802	Various	Creosote	
41	Extra Stout	Single	13.0	232817	Various	Creosote	
42	Extra Stout	Single	14.0	232836	Various	Creosote	
43	Extra Stout	Single	15.0	232944	Various	Creosote	
44	Extra Stout	Single	16.0	232925	Various	Creosote	
45	Extra Stout	Single	17.0	232812	Various	Creosote	
46	Extra Stout	Single	18.0	232813	Various	Creosote	
Stout Grade 'H' poles with fabrication drawing as stated at time of order							
47	Stout	'H'	9.0	232839	Various	Creosote	
48	Stout	'H'	10.0	233843	Various	Creosote	
49	Stout	'H'	11.0	233877	Various	Creosote	
50	Stout	'H'	12.0	233896	Various	Creosote	
51	Stout	'H'	13.0	233909	Various	Creosote	
52	Stout	'H'	14.0	233913	Various	Creosote	
53	Stout	'H'	15.0	233947	Various	Creosote	
54	Stout	'H'	16.0	233932	Various	Creosote	
55	Stout	'H'	17.0	233951	Various	Creosote	
56	Stout	'H'	18.0	233966	Various	Creosote	

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Item Ref	Grade	Single Or 'H'	Length (m)	Stock Code	Pole Fabrication Drawing Number	Preservative	Estimated Annual Usage
57	Stout	'H'	20.0	232696	Various	Creosote	
58	Stout	'H'	22.0	233985	Various	Creosote	
59	Stout	'H'	24.0	215038	Various	Creosote	
Woodhouse Mast Replacement Poles							
60	Medium	Rutter	12.0	233986	1091390008 sht2	Creosote	
61	Medium	Rutter	13.0	233987	1091390008 sht2	Creosote	
62	Medium	Rutter	14.0	233988	1091390008 sht2	Creosote	
63	Medium	Rutter	15.0	233989	1091390008 sht2	Creosote	
64	Medium	Rutter	16.0	233990	1091390008 sht2	Creosote	
65	Medium	Rutter	17.0	233991	1091390008 sht2	Creosote	
66	Stout	Rutter	12.0	233992	1091390008 sht2	Creosote	
67	Stout	Rutter	13.0	233993	1091390008 sht2	Creosote	
68	Stout	Rutter	14.0	233994	1091390008 sht2	Creosote	
69	Stout	Rutter	15.0	233995	1091390008 sht2	Creosote	
70	Stout	Rutter	16.0	233996	1091390008 sht2	Creosote	
71	Stout	Rutter	17.0	233997	1091390008 sht2	Creosote	
72	E/Stout	Single	12.0	233999	1091390003 sht2	Creosote	
73	E/Stout	Single	13.0	234000	1091390003 sht2	Creosote	
74	E/Stout	Single	14.0	234001	1091390003 sht2	Creosote	
75	E/Stout	Single	15.0	234002	1091390003 sht2	Creosote	
76	E/Stout	Single	16.0	234003	1091390003 sht2	Creosote	
77	E/Stout	Single	17.0	234004	1091390003 sht2	Creosote	
78	E/Stout	H	12.0	234005	1091390018 sht2	Creosote	
79	E/Stout	H	13.0	234006	1091390018 sht2	Creosote	
80	E/Stout	H	14.0	234007	1091390018 sht2	Creosote	
81	E/Stout	H	15.0	234008	1091390018 sht2	Creosote	
82	E/Stout	H	16.0	234009	1091390018 sht2	Creosote	
83	E/Stout	H	17.0	234010	1091390018 sht2	Creosote	
84	E/Stout	H	12.0	232639	1091390019 sht2	Creosote	
85	E/Stout	H	13.0	232643	1091390019 sht2	Creosote	
86	E/Stout	H	14.0	232658	1091390019 sht2	Creosote	
87	E/Stout	H	15.0	232662	1091390019 sht2	Creosote	
88	E/Stout	H	16.0	232728	1091390019 sht2	Creosote	
89	E/Stout	H	17.0	232677	1091390019 sht2	Creosote	
Standard Multi-fabrication Poles							
90	Stout	Single	11.0	237488	1000434010 sht 2	Creosote	
91	Stout	Single	11.0	237505	1091080864 sht3 - 1	Creosote	
92	Stout	Single	11.0	237613	1091080864 sht3 -2	Creosote	
93	Stout	Single	11.0	237098	1091070819 sht 4	Creosote	
94	Stout	'H'	11.0	237100	1091080862 sht 2	Creosote	
95	Stout	Single	9.0	237473	1091080864 sht 3	Creosote	

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Item Ref	Grade	Single Or 'H'	Length (m)	Stock Code	Pole Fabrication Drawing Number	Preservative	Estimated Annual Usage
96	Stout	Rutter	10.0	213199	Y003X3102	Creosote	
Stay and Foundation Blocks							
97				235124	439103 Type 2	CAA	
98				234668	439112 Type 1	CAA	
99				234704	439112 Type 2	CAA	
100				217497	1091010668	CAA	
101				234475	1091010650 sht 2	CAA	
102				234507	1091010650 sht 3	CAA	
103				234653	1091010650 sht2	CAA	

*** See Appendix C for copies of the drawings.

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Appendix C - Index to Fabrication Drawings Included with this specification

Standard Pole Use	Fabrication Drawing (Including Drawing Modification Letter)
<u>LV Poles</u> Combined fabricated pole for use with ABC networks - (ENA TS 43-12) and historical Open Wire Networks (ENA TS 43-30) ABC only Fabrication Drawing - ENA TS 43-12	1091193301 sht3 mod H 1091193301 sht4 mod D
<u>HV Poles - Fabricated to ENA TS 43-40</u> Single Poles 'H' Poles Single Pole Terminal (min 210mm Pole top dimension after fabrication)	1000434001 sht1, mod E 1000434002 sht1, mod D 1000434010 sht2, mod D
<u>Multi-Fab Pole Arrangements</u> Multi-Fabricated 'H' Pole Multi-Fabrication Single Terminal Pole Fabrication drawing - Single Pole Terminal Fabrication drawing - Single Pole Terminal Single Phase Totem Pole S/S 3 Phase 'H' Totem Pole S/S Short Legged 'H' pole S/S	1000434002 sht2, mod C 1000434010 sht2 1000434010 sht7, mod A 1000434010 sht8, mod A 1091070819 sht4 mod C 1091080862 sht2 mod L 1091080864 sht3 mod F
<u>EHV Poles Fabricated to</u> 66kV – OHL4 Single Pole 66kV – OHL4 'H' Pole 66/132kV – OHL9 Single Pole (Int) 66/132kV – OHL9 Single Pole (Sect/Angle) 66/132kV – OHL9 'H' Pole (Sect/Angle) 66/132kV – OHL9 'H' Pole (Term) CE/C/37 – 'H' Poles CE/C/37 – 'H' Cable Terminal Woodhouse (Rutter) Intermediate Woodhouse (Section pole) Woodhouse (Term H pole) Woodhouse (Section H pole)	1091231186 sht1 1091231187 sht1 mod J 1091380006 sht2 1091380005 sht2 1091380003 sht2 mod C 1091380004 sht2 mod A 1091380011 sht1 1091380009 sht1 1091390008 sht2 1091390003 sht2 1091390018 sht2 1091390019 sht2
Rutter Pole & Pole Key	Y003X3102 1091380007 sht1
*Stout, Extra Stout and Stout 'H' poles with various pole fabrications stated at the time of order submission	Various
Foundation Blocks ENA TS 43-91 Type 1 & 2 foundation blocks ENA TS 43-91 'H' pole foundation blocks Special 'H' pole foundation blocks Special 'H' pole foundation blocks CE/C/37 Brace Blocks	1000439103 sht1, mod E 1000439112 sht2, mod D 1091010650 sht2 mod C 1091010650 sht3 mod E 1091010668 sht1

* Note

The Company has a requirement for a range of additional pole fabrications, to cater for in house company design standards or to support special pole mounted plant. The Company will submit copies of these pole fabrication drawings at the time of order placement.

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Appendix D - Addendum to Supplier Requirements

Bleeding Poles

When poles are identified as defective due to bleeding, the Company will notify the Supplier at the earliest opportunity. Following notification the Supplier will, without prejudice to or payment from the Company, arrange to collect, clean and return those poles to the Company. This work must be carried out within one week of the Supplier receiving notification from the Company.

Species of wood utilised for Poles

Poles should only be obtained from sustainable forests. Tenderers should state the country of origin of poles and provide verification statements confirming the sustainable management of source forests. The successful tenderer will be required to provide annual statistics detailing:

- Countries of pole origin
- Numbers of poles from each country
- Details of source forest sustainability management schemes.

Water Soluble Preservative Treated Poles

Suppliers are requested to provide information, including a safety data sheet, for the proposed preservation treatment.

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Appendix E – Logistical Requirements

Disposal of used poles

Tenderers shall indicate the services they provide to collect and dispose of used wood poles and blocks.

Stockholding

The tenderers shall provide details of their pole stocks that would be available to the company during abnormal occurrences and also details of their potential delivery times during these abnormal occurrences